

KUNAL HEMANT PASAD

Software Engineer | Distributed Systems | Backend Engineering | AI-Powered Applications
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EDUCATION

Rutgers, The State University of New Jersey

Master of Science in Computer Science | GPA: 3.83/4.0

Sardar Patel Institute of Technology

Bachelor of Technology in Computer Engineering | GPA: 8.62/10

New Brunswick, NJ

Expected May 2027

Mumbai, India

Aug 2021 – May 2025

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, C++, OOP

Backend & Frameworks: Spring Boot, FastAPI, Node.js, REST APIs, GraphQL, Microservices, Kafka Streams, Celery, System Design

AI & Machine Learning: LangGraph, LangChain, RAG, Vector Search, pgvector, Fine-Tuning, QLoRA, CatBoost, OpenAI API

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis, TimescaleDB, RocksDB

Cloud, DevOps & Tools: AWS (SQS, S3, CloudWatch, Glacier), Docker, Kubernetes, GitHub Actions, CI/CD, OpenTelemetry

PROFESSIONAL EXPERIENCE

National Appraisal Partners

Jun 2026 – Present

AI Automation Intern

Texas, USA (Remote)

- Built an end-to-end ML system predicting real-estate appraisal fees and recommending appraisers from raw bid inputs, replacing a fully manual pricing workflow
- Engineered a data pipeline normalizing 5,000+ historical bids from FileMaker Pro with shared preprocessing that guaranteed exact train/serve parity across all 364 input fields
- Deployed trained models as a Flask REST API integrated into Google Sheets via Apps Script, enabling one-click predictions for the bidding team

EternIQ

Jan 2025 – May 2025

Software Engineer Intern

Mumbai, India

- Built scalable LLM inference pipelines using chain-of-thought prompting and structured outputs, reducing hallucination rates by approximately 25% across 1K+ evaluated responses
- Developed a model evaluation framework to track reasoning accuracy and failure modes, establishing reproducible quality benchmarks with unit testing across deployment iterations
- Built data ingestion pipelines processing 100K+ records with schema validation, normalization, and REST API integration into production systems in an agile development environment

Mukesoft IT Consultants

Jul 2023 – Feb 2024

Software Developer Intern

Mumbai, India

- Built backend REST APIs using Laravel and CodeIgniter; optimized SQL queries with composite indexing, reducing API response latency by approximately 30% in production
- Processed 50K+ relational records with schema validation and integrity checks, improving data consistency and reliability across the application

PROJECTS

PulseQueue | Java, Spring Boot, Kafka Streams, RocksDB, AWS SQS, S3, Docker, React | [GitHub](#) | [Live](#)

Nov 2025

- Built a distributed event-driven streaming and observability platform that computes p50/p95/p99 latency in 1-minute tumbling windows using Kafka Streams and a RocksDB state store, with results streamed live to a React dashboard over SSE
- Published a zero-config Spring Boot SDK that auto-instruments any Spring MVC app via a servlet filter; implemented Welford anomaly detection and fault-tolerant AWS SQS dead-letter recovery for failed Kafka events
- Achieved p95 = 148ms across 8,660 requests with zero failures in Gatling load testing; configured GitHub Actions CI/CD with Docker builds and monitoring on every commit

Synapse | LangGraph, FastAPI, Celery, Redis, pgvector, OpenTelemetry, React, Docker | [GitHub](#) | [Live](#)

Mar 2026

- Architected a multi-agent orchestration platform using LangGraph where a supervisor agent decomposes and routes tasks across five specialist agents, achieving 76% success on the GAIA Level 1 benchmark, 18 points above GPT-4o
- Designed two-tier memory using Redis for session state and pgvector for long-term episodic recall with MMR re-ranking, maintaining sub-15ms p99 retrieval latency across 100K stored chunks
- Built a real-time WebSocket agent trace viewer with OpenTelemetry spans and used Celery with dead-letter recovery to achieve zero task loss across 10,000 simulated runs

InferMesh | FastAPI, PostgreSQL, Redis, pgvector, Celery, React, Docker | [GitHub](#) | [Live](#)

Jan 2026

- Built a scalable self-hosted LLM inference gateway for Gemini, OpenAI, and Anthropic with automatic cost-based provider routing, JWT authentication, and per-request cost tracking
- Implemented semantic caching via pgvector cosine similarity on 3072-dim embeddings; built an async Celery quality eval pipeline and Redis sliding-window rate limiting at 60 req/min per user
- Engineered Fernet-symmetric API key encryption for per-user provider credential isolation and implemented JWT-based auth with email verification, making the platform safe for multi-tenant production use

PUBLICATIONS

CultSportQA: A Multilingual QA Benchmark for Traditional Sports | [Paper](#)

EMNLP 2025 (Main Conference) | Co-author

- Co-authored a multilingual QA benchmark of 33K+ questions across 60 countries and 11 languages; evaluated 17+ LLMs under zero-shot, few-shot, and chain-of-thought prompting